



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Closure properties of context-free languages

Dr Noleen Koehler

University of Leeds

Closure properties of context-free languages

Operations on languages

Definition

Let A and B be two languages over Σ , we define the following operations,

- **Union:** $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- **Complement:** $\overline{A} = \{x \in \Sigma^* \mid x \notin A\}$
- **Concatenation:** $A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$
- **Kleene star:**
 $A^* = \{x_1 \dots x_k \mid k \geq 0 \text{ and, for each } 0 \leq i \leq k, x_i \in A\}$

Theorem (Closure with respect to Union)

Context-free languages are closed with respect to Union.

Theorem

Context-free languages are not closed with respect to Intersection.

Theorem

Context-free languages are not closed with respect to Complement.

Theorem (Closure with respect to Intersection with a regular language)

Context-free languages are closed with respect to Intersection with a regular language.

Theorem (Closure with respect to Concatenation)

Context-free languages are closed with respect to Concatenation.

Theorem (Closure with respect to Kleene star)

Context-free languages are closed with respect to Kleene star.